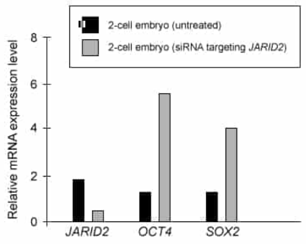
The graph below shows messenger RNA (mRNA) expression levels for JARID2, a protein known to regulate gene expression during embryonic development, and two of its target genes (*OCT4* and *SOX2*) in cattle (*Bos taurus*) embryos at the two-cell stage. These embryos had been either treated with small interfering RNA (siRNA) targeting *JARID2* mRNA or left untreated.

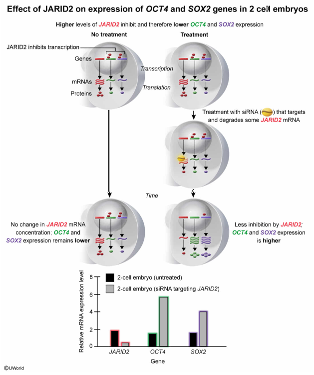


Which of the following most likely explains why expression of *OCT4* and *SOX2* mRNA was higher in the siRNA-treated embryos?

- A. *OCT4* and *SOX2* are essential for *JARID2* gene expression. [2%]
- B. Translation of *OCT4* and *SOX2* occurs more rapidly when *JARID2* gene expression is lower. [12%]
- ✔ C. JARID2 normally decreases *OCT4* and *SOX2* transcription at the two-cell stage in *B. taurus* embryos. [73%]
- D. JARID2 normally increases *OCT4* and *SOX2* transcription at the two-cell stage in *B. taurus* embryos. [10%]

Correct
74% answered correctly

Explanation:



During embryonic development, gene expression among cells is regulated by multiple factors (eg, morphogen concentration, transcription factors). Gene regulation allows cells in a developing organism, which all contain the same genome, to differentiate and express different sets of genes.

Due to this regulation of gene expression, the amount of messenger RNA (mRNA) transcribed from certain genes varies depending on cell type and developmental stage. Differences in mRNA expression in turn cause variations in the amount and type of translated proteins, leading to phenotypic and functional differentiation among cell types during development. In addition, the degradation of certain mRNA molecules during different developmental stages helps ensure regulation of expression timing for encoded proteins.

The graph in the question shows mRNA expression levels for JARID2, a protein known to regulate gene expression during embryonic development. The graph also shows mRNA expression levels for two JARID2 target genes (*OCT4* and *SOX2*) in two-cell *Bos taurus* embryos that had been either left untreated or treated with *JARID2*-targeting small interfering RNA (siRNA) (ie, small RNA molecules that cause mRNA degradation).

According to the data, **decreased** *JARID2* mRNA expression in siRNA-treated embryos led to **increased** *OCT4* and *SOX2* mRNA expression compared to untreated embryos. This indicates that **JARID2 normally decreases** (*not* increases) ***OCT4* and *SOX2* transcription** at the two-cell stage in *B. taurus* embryos **(Choice D)**. However, expression of *OCT4* and *SOX2* can increase when the amount of JARID2 is *reduced* via siRNA treatment.

(Choice A) The experiment in the question does *not* provide information on how the presence or absence of *OCT4* and *SOX2* in cells affects *JARID2* gene expression. Accordingly, to determine if *OCT4* and *SOX2* are essential for *JARID2* expression, an experiment should be performed that compares *JARID2* expression in untreated (control) cells and in cells in which expression of *OCT4* and *SOX2* is prevented.

(Choice B) Although *OCT4* and *SOX2* mRNA expression (ie, transcription) *increased* when *JARID2* expression was lower, the data in the graph do *not* provide information on rates of *OCT4* and *SOX2 translation*.

Educational objective:
Gene expression (ie, messenger RNA and protein expression) among cells is regulated during embryonic development. As an organism develops, this regulation of gene expression allows cells to differentiate and express unique sets of genes conferring specific functions.

Time spent: 0 sec
Subject: Biology

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System: Reproduction